SystemUI

- hurricaneReliefSystem: HurricaneReliefSystem
- carterRequest: HelpRequest

+ HurricaneReliefSystemUI()
+ run(): void
+ main(String[] args): void
+ createRequest(): void
+ matchRequest(): void
+ findShelter(): void
+ exportTripInfo(): void
+ permissionsCheck(): void

TripExporter

+ exportTripInfo(HelpRequest request, Volunteer volunteer, Shelter shelter, String fileName): void
- getFamilyMemberInfo(User user): ArrayList<String>

User

- userID: UUID
- type: String
- firstName: String
- lastName: String
- username: String
- password: String
- email: String
- phoneNumber: String
- dob: Date
- location: Location
- emergencyContacts: ArrayList<EmergencyContact>
- medicalInfo: ArrayList<String>

+ User(UUID userID, String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location, ArrayList<EmergencyContact> emergencyContacts, ArrayList<String> medicalInfo)
+ User(String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location, ArrayList<EmergencyContact> emergencyContacts, ArrayList<String> medicalInfo)
+ changeUsername(String username): void
+ changePassword(String password): void
+ exists(String username, String password): boolean

EmergencyContact

- firstName: String
- lastName: String
- age: int
- email: String
- phoneNumber: String
- relationship: String

+ EmergencyContact(String firstName, String lastName, int age, String email, String phoneNumber, String relationship)

Guest

- phoneNumber: String
- location: Location

+ Guest(String phoneNumber, Location location)
+ createRequest(String description, int numberOfPeople, int numberOfPets, String petInfo, Urgency urgency): HelpRequest
+ updateRequest(HelpRequest request, String description, Location location, int numberOfPeople, int numberOfPets, Urgency urgency): void
+ cancelRequest(HelpRequest request): void

Location

- address: String
- city: String
- state: String
- zipCode: String

+ Location(String address, String city, String state, String zipCode)

HurricaneReliefSystem

- hurricaneReliefSystem: HurricaneReliefSystem
- userList: UserList
- requestList: RequestList
- shelterList: ShelterList
- currentUser: User
- currentRequest: HelpRequest
- currentShelter: Shelter
- hurricaneData: HurricaneData

- HurricaneReliefSystem()
+ getInstance(): HurricaneReliefSystem
+ findRequest(): ArrayList<HelpRequest>
+ findRequest(Location location): ArrayList<HelpRequest>
+ findRequest(Location location, boolean requiresCertification): ArrayList<HelpRequest>
+ findRequest(Location location, boolean requiresCertification, Skill skill): ArrayList<HelpRequest>
+ findAvailableRequests(Volunteer volunteer): ArrayList<HelpRequest>
+ getRecommendedRequests(Volunteer volunteer): ArrayList<HelpRequest>
+ getRecommendationExplanation(Volunteer volunteer, HelpRequest request): String
+ formatRecommendedRequest(Volunteer volunteer, HelpRequest request): String
- getRecommendationScore(Volunteer volunteer, HelpRequest request): int
- isNearby(Location firstLocation, Location secondLocation): boolean
+ findShelter(String name): ArrayList<Shelter>
+ findShelter(String name, Location location): ArrayList<Shelter>
+ findShelter(String name, Location location, boolean allowsPets): ArrayList<Shelter>
+ findShelter(boolean allowsPets, boolean requiresMedical, int minCapacity): ArrayList<Shelter>
+ findShelterByDrivingDistance(Location location, ArrayList<Shelter> shelters): Shelter
+ createAccount(String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location): User
+ login(String username, String password): User
+ logout(): void
+ getCurrentUser(): User
+ getCurrentShelter(): Shelter
+ getCurrentRequest(): HelpRequest
+ createLocation(String address, String city, String state, String zipCode): Location
+ cancelRequest(User user, HelpRequest request): boolean
+ addShelter(String name, Location location, int maxCapacity, int currentOccupancy, boolean allowsPets, boolean isOpen, ArrayList<Resource> resources): boolean
+ verifyAdmin(User user): boolean
+ addRequest(User author, String description, Location location, boolean requiresCertification, ArrayList<Skill> neededSkills, int numberOfPeople, int numberOfPets, String petInfo, Status status, Urgency urgency): boolean
+ addEmergencyContact(EmergencyContact emergencyContact): boolean
+ updateRequestStatus(User user, HelpRequest request): boolean
+ checkInFamily(Shelter shelter, ArrayList<String> memberNames, String pet): boolean
+ completeRequest(User user, HelpRequest request): boolean
+ comment(HelpRequest request, User author, String timestamp, String text): boolean
+ viewHurricaneData(): HurricaneData
+ updateHurricaneData(HurricaneData hurricaneData): void
+ volunteer(User user, HelpRequest request): boolean
+ saveUsers(): void
+ saveShelters(): void
+ saveRequests(): void

UserList

- userList: UserList
- users: ArrayList<User>

- UserList()
+ getInstance(): UserList
+ getUsers(): ArrayList<User>
+ getUser(String username): User
+ getUser(String username, String password): User
+ addUser(User user): boolean
+ createAccount(String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location): User
+ save(): void

Victim

- activeRequests: ArrayList<HelpRequest>

+ Victim(UUID userID, String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location, ArrayList<EmergencyContact> emergencyContacts, ArrayList<String> medicalInfo)
+ cancelRequest(HelpRequest request): void
+ submitRequestForOther(User user, HelpRequest request): void
+ addActiveRequest(HelpRequest request): void

Volunteer

- skills: ArrayList<Skill>
- certifications: ArrayList<String>
- hasVehicle: boolean
- assignedRequests: ArrayList<HelpRequest>
- isAvailable: boolean
- vehicleDescription: String
- approved: boolean

+ Volunteer(UUID userID, String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location, ArrayList<EmergencyContact> emergencyContacts, ArrayList<String> medicalInfo, boolean hasVehicle)
+ addSkill(Skill skill): void
+ acceptRequest(HelpRequest request): void
+ releaseRequest(HelpRequest request): void
+ updateRequestStatus(HelpRequest request): void
+ hasSkill(Skill skill): boolean
+ addCertification(String certification): void
+ canAcceptRequest(HelpRequest request): boolean
+ restoreAssignment(HelpRequest request): void

SystemAdmin

- sheltersManaged: ArrayList<Shelter>

+ SystemAdmin(UUID userID, String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location, ArrayList<EmergencyContact> emergencyContacts, ArrayList<String> medicalInfo)
+ approveVolunteer(Volunteer volunteer): void
+ addShelter(String name, Location location, int maxCapacity, int currentOccupancy, boolean allowsPets, boolean isOpen, ArrayList<Resource> resources): boolean
+ removeManagedShelter(Shelter shelter): void
+ removeRequest(HelpRequest request): void
+ updateShelterInfo(Shelter shelter): void
+ updateHurricaneData(HurricaneData hurricaneData): void

ShelterManager

- assignedShelter: Shelter

+ ShelterManager(UUID userId, String type, String firstName, String lastName, String username, String password, String email, String phoneNumber, Date dob, Location location, ArrayList<EmergencyContact> emergencyContacts, ArrayList<String> medicalInfo, Shelter assignedShelter)
+ getAssignedShelter(): Shelter
+ setAssignedShelter(Shelter shelter): void

RequestList

- requestList: RequestList
- requests: ArrayList<HelpRequest>

- RequestList()
+ getInstance(): RequestList
+ getRequests(): ArrayList<HelpRequest>
+ getRequests(Location location): ArrayList<HelpRequest>
+ getRequests(Location location, boolean requiresCertification): ArrayList<HelpRequest>
+ getRequests(Location location, boolean requiresCertification, Skill skill): ArrayList<HelpRequest>
+ addRequest(User author, String description, Location location, boolean requiresCertification, ArrayList<Skill> neededSkills, int numberOfPeople, int numberOfPets, String petInfo, Status status, Urgency urgency): boolean
+ addRequest(HelpRequest request): boolean
+ removeRequest(HelpRequest request): boolean
+ save(): void

ShelterList

- shelterList: ShelterList
- shelters: ArrayList<Shelter>

- ShelterList()
+ getInstance(): ShelterList
+ getShelters(): ArrayList<Shelter>
+ getShelters(String name): ArrayList<Shelter>
+ getShelters(String name, Location location): ArrayList<Shelter>
+ getShelters(String name, Location location, boolean allowsPets): ArrayList<Shelter>
+ getShelters(boolean allowsPets, boolean hasMedicalStaff, int minAvailableSpaces): ArrayList<Shelter>
+ findShelterByDrivingDistance(Location location, ArrayList<Shelter> shelters): Shelter
+ addShelter(Shelter shelter): boolean
+ removeShelter(Shelter shelter): boolean
+ clearShelters(): void
+ addShelter(String name, Location location, int maxCapacity, int currentOccupancy, boolean allowsPets, boolean isOpen, ArrayList<Resource> resources): boolean
+ save(): void
- matchesName(Shelter shelter, String name): boolean
- matchesLocation(Shelter shelter, Location location): boolean
- allowPetsMatch(Shelter shelter, boolean allowsPets): boolean
- hasMedicalStaffMatch(Shelter shelter, boolean hasMedicalStaff): boolean
- hasSpace(Shelter shelter, int minAvailableSpaces): boolean

Shelter

- shelterID: UUID
- name: String
- location: Location
- maxCapacity: int
- currentOccupancy: int
- allowsPets: boolean
- isOpen: boolean
- resources: ArrayList<Resource>
- occupants: ArrayList<String>
- pets: ArrayList<String>
- hasMedicalStaff: boolean

+ Shelter(UUID shelterID, String name, Location location, int maxCapacity, int currentOccupancy, boolean allowsPets, boolean isOpen, ArrayList<Resource> resources)
+ Shelter(String name, Location location, int maxCapacity, int currentOccupancy, boolean allowsPets, boolean isOpen, ArrayList<Resource> resources)
+ Shelter(UUID shelterID, String name, Location location, int maxCapacity, int currentOccupancy, boolean allowsPets, boolean isOpen, ArrayList<Resource> resources, ArrayList<String> occupants, ArrayList<String> pets, boolean hasMedicalStaff)
+ Shelter(String name, Location location, int maxCapacity, int currentOccupancy, boolean allowsPets, boolean isOpen, ArrayList<Resource> resources, ArrayList<String> occupants, ArrayList<String> pets, boolean hasMedicalStaff)
+ hasVacancy(): boolean
+ getAvailableResources(): ArrayList<Resource>
+ checkInUser(User user): void
+ updateOperationalStatus(): void
+ checkInFamily(ArrayList<String> memberNames, String petNameAndType): boolean

DataConstants

USER CONSTANTS
REQUEST CONSTANTS
SHELTER CONSTANTS

DataLoader

+ loadUsers(): ArrayList<User>
+ loadShelters(): ArrayList<Shelter>
+ loadRequests(): ArrayList<HelpRequest>

DataWriter

+ saveUsers(): void
+ saveShelters(): void
+ saveRequests(): void

HelpRequest

- requestID: UUID
- author: User
- description: String
- location: Location
- requiresCertification: boolean
- neededSkills: ArrayList<Skill>
- numberOfPeople: int
- numberOfPets: int
- status: Status
- urgency: Urgency
- hurricaneData: HurricaneData
- commentThread: ArrayList<Comment>
- petInfo: String
- guestPhoneNumber: String
- assignedVolunteer: User
- selectedShelter: Shelter
- requiresMedicalStaff: boolean

+ HelpRequest(UUID requestID, User author, String description, Location location, boolean requiresCertification, ArrayList<Skill> neededSkills, int numberOfPeople, int numberOfPets, String petInfo, Status status, Urgency urgency)
+ HelpRequest(User author, String description, Location location, boolean requiresCertification, ArrayList<Skill> neededSkills, int numberOfPeople, int numberOfPets, String petInfo, Status status, Urgency urgency)
+ addComment(Comment comment): void
+ updateStatus(): void
+ isARequest(Location location, boolean requiresCertification, Skill skill): boolean
+ isActiveAssignment(): boolean
+ isTerminal(): boolean

<<enumeration>> Skill

EMT
CPR
MEDICAL
POLICE
CONSTRUCTION
TRANSPORTATION

<<enumeration>> Status

OPEN
ASSIGNED
IN_PROGRESS
COMPLETE
CANCELLED

<<enumeration>> Urgency

CRITICAL
HIGH
MEDIUM
LOW

Resource

- name: String
- quantity: int
- unit: String

+ Resource(String name, int quantity, String unit)

HurricaneData

- windSpeeds: int
- category: int

+ HurricaneData(int windSpeeds, int category)
+ updateData(int windSpeeds, int category): void

Comment

- author: User
- timestamp: String
- text: String
- replies: ArrayList<String>

+ Comment(User author, String timestamp, String text)
+ addReply(String reply): void